

Objective

The purpose of this activity was to analyze the lab network, identify active hosts, enumerate services, capture network traffic, and evaluate security implications within an isolated testing environment.

Testing was conducted against systems owned and controlled by the project author within a virtual lab environment.

Lab Environment

Network Components

Component	IP Address	Role
Parrot Security VM	192.168.64.3	Testing workstation
macOS Host	192.168.64.1	Gateway and Docker host
OWASP Juice Shop	192.168.64.1:3000	Vulnerable web application
Network	192.168.64.0/24	Isolated lab subnet

```

[user@parrot]~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:27:43:47 brd ff:ff:ff:ff:ff:ff
    altname enx525400274347
    inet 192.168.64.3/24 brd 192.168.64.255 scope global dynamic noprefixroute enp0s1
        valid_lft 3591sec preferred_lft 3591sec
    inet6 fd1b:e2f0:f39e:1a01:78fb:3d3d:d1fc:1440/64 scope global dynamic noprefixroute
        valid_lft 2591992sec preferred_lft 604792sec
    inet6 fe80::4ec:2c:698d:8a4f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[user@parrot]~$ ip route
default via 192.168.64.1 dev enp0s1 proto dhcp src 192.168.64.3 metric 100
192.168.64.0/24 dev enp0s1 proto kernel scope link src 192.168.64.3 metric 100
[user@parrot]~$ nmap -sn 192.168.64.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-30 15:37 UTC
Nmap scan report for 192.168.64.1
Host is up (0.0016s latency).
Nmap scan report for 192.168.64.3
Host is up (0.00040s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 3.09 seconds

```

Analysis

The Parrot VM operated on the 192.168.64.0/24 subnet using interface `enp0s1`. The default gateway and Docker host were identified as 192.168.64.1. OWASP Juice Shop was exposed through TCP port 3000 on the host system.

Network Discovery

Host discovery was performed using Nmap ICMP and ARP scanning.

Results

IP Address	Status	Role
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192.168.6 Up macOS Host /
4.1 Gateway

192.168.6 Up Parrot VM
4.3

```
[user@parrot]~$ nmap -sn 192.168.64.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-01 15:44 UTC
Nmap scan report for 192.168.64.1
Host is up (0.0011s latency).
Nmap scan report for 192.168.64.3
Host is up (0.00016s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 2.96 seconds
```

Analysis

Only two active systems were identified within the isolated lab environment:

- Testing VM
- Host system

This confirms a simple network topology suitable for controlled security testing.

Service Enumeration

TCP service enumeration was conducted to identify exposed services.

```
[x]-[user@parrot]-[~]  
$nmap -p 3000 --script http-title,http-headers 192.168.64.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-30 18:07 UTC  
Nmap scan report for 192.168.64.1  
Host is up (0.00034s latency).  
  
PORT      STATE SERVICE  
3000/tcp  open  ppp  
  
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

Analysis

Port 3000 was identified as open and associated with the OWASP Juice Shop web application.

Nmap labeled the service as:

3000/tcp open ppp

However, packet analysis confirmed the service was actually HTTP.

UDP Enumeration

UDP testing was performed to identify additional network services.

```

[user@parrot]-[~]
$ sudo nmap -sU --top-ports 20 192.168.64.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-30 18:19 UTC
Nmap scan report for 192.168.64.1
Host is up (0.00029s latency).

PORT      STATE      SERVICE
53/udp    open       domain
67/udp    open|filtered dhcp
68/udp    closed     dhcp
69/udp    closed     tftp
123/udp   closed     ntp
135/udp   closed     msrpc
137/udp   open       netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   closed     netbios-ssn
161/udp   closed     snmp
162/udp   closed     snmptrap
445/udp   closed     microsoft-ds
500/udp   closed     isakmp
514/udp   closed     syslog
520/udp   closed     route
631/udp   closed     ipp
1434/udp  closed     ms-sql-m
1900/udp  closed     upnp
4500/udp  closed     nat-t-ike
49152/udp closed     unknown
MAC Address: DE:93:96:D0:90:64 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 1.33 seconds

```

Port	Service	Status
53	DNS	Open
67	DHCP	Open/Filtered
137	NetBIOS	Open

138	NetBIOS Datagram	Open/Filtered
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Analysis

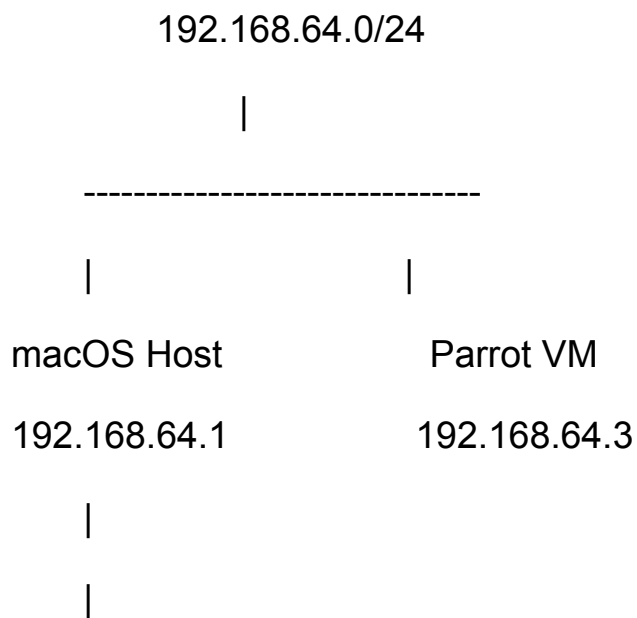
Several UDP services were identified:

- DNS (53)
- DHCP (67)
- NetBIOS (137/138)

These services support:

- name resolution
- address assignment
- network sharing

Open UDP services represent additional network exposure and should be evaluated in production environments.



Docker Engine

|

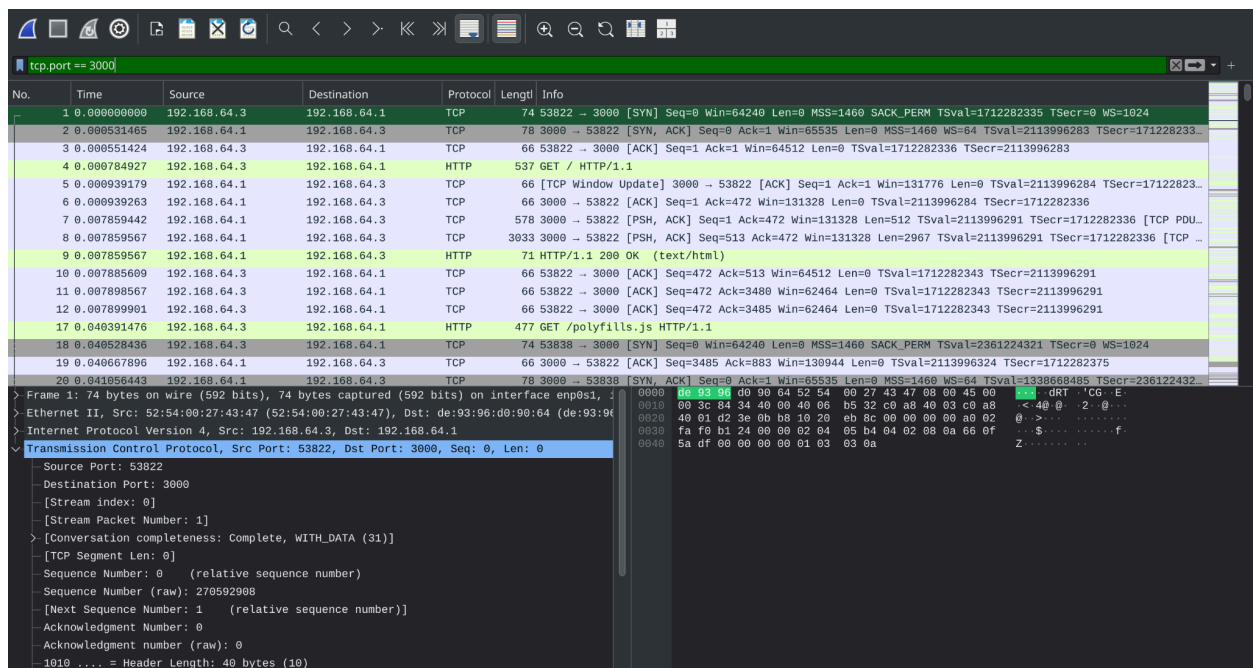
|

OWASP Juice Shop

TCP/3000

Traffic Analysis

Traffic analysis was performed using Wireshark.



TCP Three-Way Handshake

Observed:

192.168.64.3 → 192.168.64.1 SYN

192.168.64.1 → 192.168.64.3 SYN, ACK

192.168.64.3 → 192.168.64.1 ACK

Analysis

This demonstrates the TCP connection establishment process prior to application communication.

This directly shows understanding of:

- TCP
- IP
- Transport-layer protocols

HTTP Analysis

HTTP communication was observed after connection establishment.

Request:

GET / HTTP/1.1

Host: 192.168.64.1:3000

Response:

HTTP/1.1 200 OK

Content-Type: text/html

Security Headers Observed

Header

Purpose

X-Content-Type-Options	Prevent MIME attacks
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X-Frame-Options	Clickjacking protection
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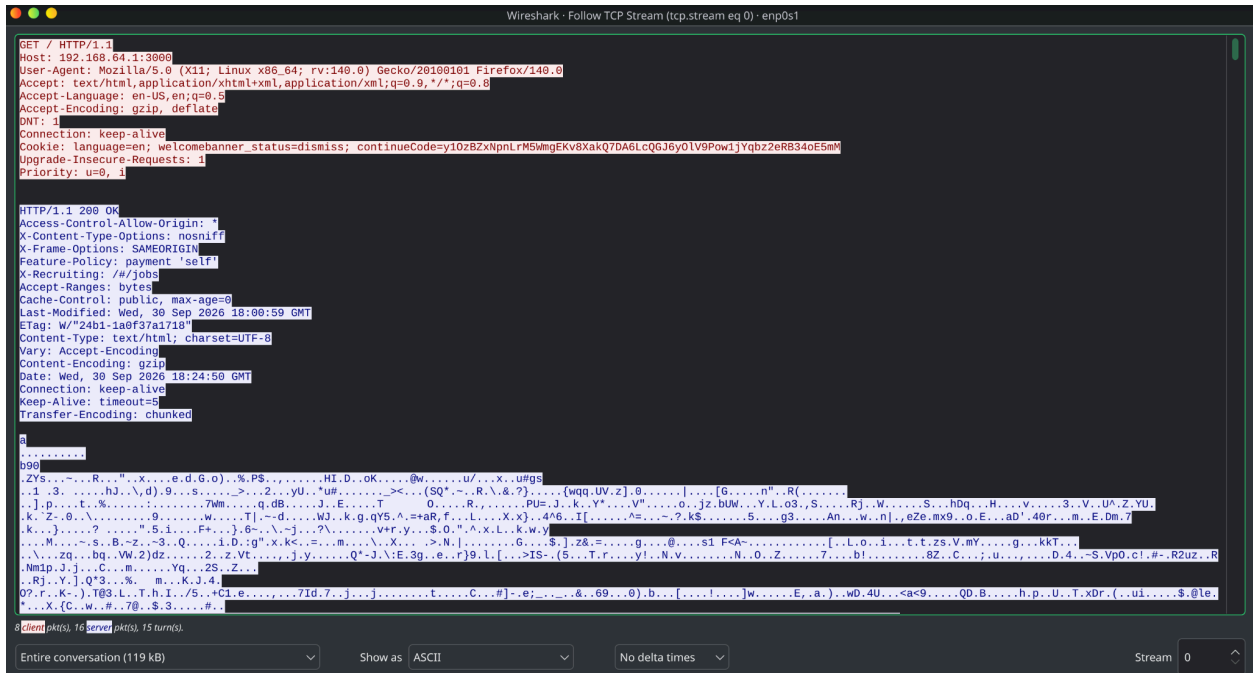
Feature-Policy	Browser restrictions
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Analysis

Traffic analysis confirmed:

- HTTP communication
- server responses
- security headers
- application behavior

Because traffic was transmitted using HTTP rather than HTTPS, communication could be observed in clear text.



Access Points

The following access points were identified:

Access Point	Description
TCP/3000	Web application access
UDP/53	DNS
UDP/67	DHCP

UDP/137 NetBIOS

Security Implications

Open services increase network exposure.

Potential risks include:

- service exploitation
- information disclosure
- unauthorized access
- service enumeration

Findings

NT-001 — Network Accessible Web Application

Evidence:

- TCP/3000 open
- HTTP traffic captured

Risk:

The application is reachable over the network and could be targeted if vulnerabilities exist.

NT-002 — Unencrypted Web Traffic

Evidence:

- Wireshark captured readable HTTP traffic.

Risk:

Sensitive information transmitted over HTTP may be visible to network observers.

NT-003 — Additional UDP Services**Evidence:**

- DNS
- DHCP
- NetBIOS

Risk:

Unnecessary UDP services may expand attack surface.

Recommendations

1. Limit exposed services.
2. Use firewalls and segmentation.
3. Replace HTTP with HTTPS in production.
4. Monitor open UDP services.
5. Regularly perform service enumeration.
6. Validate services using packet analysis.